

Third Normal Form (3NF) Relational Database Design

> **Definition:** A relation is in **Third Normal Form (3NF)** if it is in **2NF** and **no non-prime attribute is transitively dependent on the primary key**. For every non-trivial functional dependency $X \rightarrow Y$, either **X is a super key** or **Y is a prime attribute**.

1. Detailed Technical Explanation

Transitive Dependency Concept:

A transitive dependency exists if $A \rightarrow B$ and $B \rightarrow C$ hold, which implies $A \rightarrow C$ through non-key attribute B.

...

Conversion Example: 2NF to 3NF

Relation EMP_DEPT(Emp_ID, Emp_Name, Dept_ID, Dept_Name, Dept_Head)

Decomposition into 3NF:

1. **EMPLOYEE (Emp_ID, Emp_Name, Dept_ID)**

2. Core Concepts & Memory Keywords

- **Transitive Dependency:** Non-key attribute determining another non-key attribute.

3. Must-Write Points for Exams

- 3NF guarantees **lossless-join decomposition** and **dependency preservation**.

4. Quick Recall Flow

2NF R